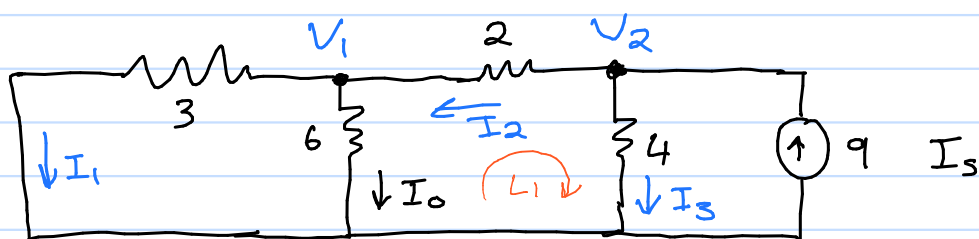


4.4 Linearity property

Using linearity determine I_o



$$\text{Assume } I_o = 1 \text{ A} \quad (\text{Linearity})$$

$$V_1 = 6 I_o = 6 \text{ V} \quad (\text{Ohm's law})$$

$$I_1 = \frac{V_1}{3} = \frac{6}{3} = 2 \text{ A} \quad (\text{Ohm's law})$$

$$\begin{aligned} I_2 &= I_1 + I_o \\ &= 2 + 1 \\ &= 3 \text{ A} \end{aligned} \quad (\text{KCL at } V_1)$$

$$\begin{aligned} &+ - 2 I_2 + + \\ &V_1 \quad \quad V_2 \\ &- \quad \quad - \end{aligned} \quad (\text{KVL at } L_1)$$

$$-V_1 - 2 I_2 + V_2 = 0$$

$$\begin{aligned} V_2 &= V_1 + 2 I_2 \\ &= 6 + 2 \times 3 \\ &= 12 \text{ V} \end{aligned}$$

$$I_3 = \frac{V_2}{4} \quad (\text{Ohm's law})$$

$$= \frac{12}{4}$$

$$= 3 \text{ A}$$

$$I_s = I_2 + I_3 \quad (\text{KCL at } V_2)$$

$$\begin{aligned} &= 3 + 3 \\ &= 6 \text{ A} \end{aligned}$$

$$\text{This gives } I_s = 6 I_o \quad (\text{Linearity})$$

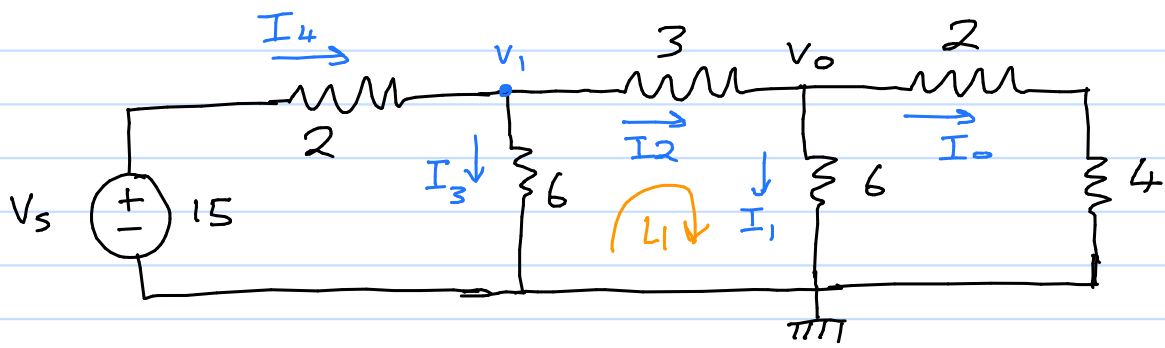
$$I_o = \frac{I_s}{6}$$

$$= \frac{9}{6}$$

$$= 1.5 \text{ A}$$

4.5 Linearity property

Assume $V_o = 1\text{ V}$, use linearity to find V_o



Assume $V_o = 1\text{ V}$

$$I_o = \frac{1}{6} \quad I_1 = \frac{1}{6} \quad (\text{Ohm's law})$$

$$I_2 = I_1 + I_o = \frac{2}{6} \quad (\text{KCL at } V_o)$$

$$\begin{aligned} V_1 &= V_o + 3I_2 \\ &= 1 + 3 \times \frac{2}{6} \\ &= 2\text{ V} \end{aligned} \quad (\text{KVL at } L_1)$$

$$I_3 = \frac{2}{6}$$

$$\begin{aligned} I_4 &= I_3 + I_2 \\ &= \frac{2}{6} + \frac{2}{6} \\ &= \frac{4}{6} \end{aligned} \quad (\text{KCL at } V_1)$$

$$I_4 = \frac{V_s - V_1}{2} \quad (\text{Ohm's law})$$

$$\frac{4}{6} = \frac{V_s - 2}{2}$$

$$4 = 3V_s - 6$$

$$V_s = \frac{10}{3}$$

$$V_s = \frac{10}{3} V_o \quad (\text{Linearity})$$

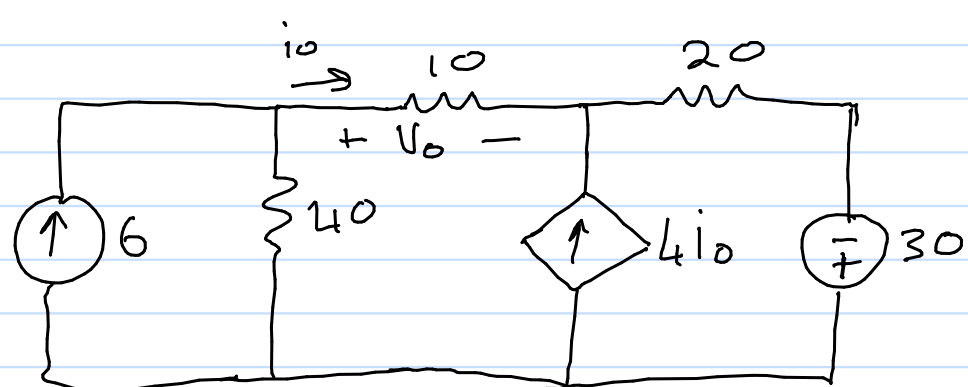
$$V_o = \frac{V_s \cdot 3}{10}$$

$$= \frac{15 \times 3}{10}$$

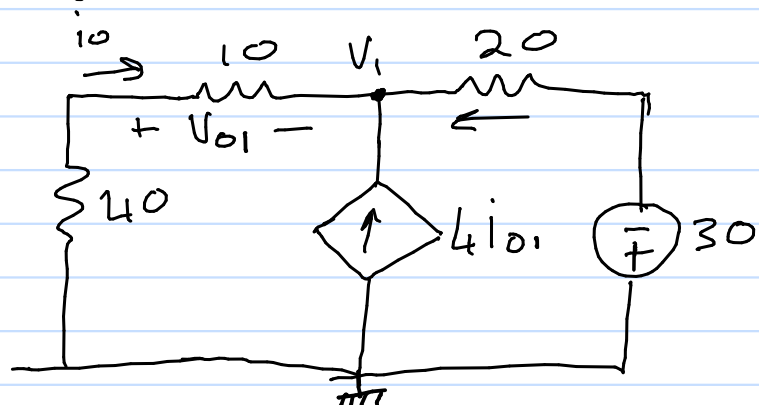
$$= 4.5\text{ V}$$

4.11 Superposition

Use superposition to find V_o & i_o



Switch of 6A source



KCL at V_1

$$i_{o1} + 4i_{o1} + \frac{(-30) - V_1}{20} = 0 \quad (1)$$

Ohm's law at V_1

$$V_1 = -(40 + 10)i_{o1} = -50i_{o1} \quad (2)$$

(Voltage always drops when current flows through resistor, V_1 is a drop from ground)

Substitute 2 into 1

$$5i_o + \frac{(-30) + 50i_{o1}}{20} = 0$$

$$100i_o - 30 + 50i_{o1} = 0$$

$$150i_{o1} = 30$$

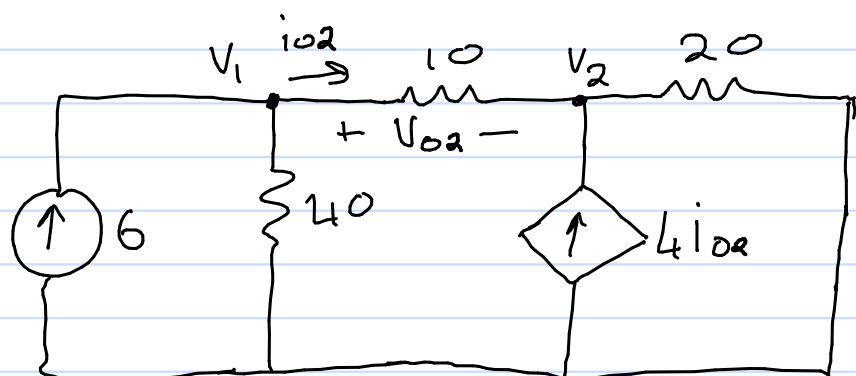
$$i_{o1} = 0,2 \text{ A}$$

Ohm's law at V_o

$$V_{o1} = 10i_{o1}$$

$$= 2 \text{ V}$$

Switch of 30 V source



$$V_o = V_1 - V_2$$

Perform nodal analysis

$$-6 + \frac{V_1}{40} + \frac{V_1 - V_2}{10} = 0$$

$$V_1 + 4V_1 - 4V_2 = 240$$

$$5V_1 - 4V_2 = 240 \quad (1)$$

$$\frac{V_2 - V_1}{10} - 4i_o + \frac{V_2}{20} = 0$$

$$\text{substitute } i_o = \frac{V_1 - V_2}{10}$$

$$\frac{V_2 - V_1}{10} - \frac{4(V_1 - V_2)}{10} + \frac{V_2}{20} = 0$$

$$2V_2 - 2V_1 - 8V_1 + 8V_2 + V_2 = 0$$

$$-10V_1 + 11V_2 = 0 \quad (2)$$

from 1 and 2

$$V_1 = 176 \text{ V} \quad V_2 = 160 \text{ V}$$

$$V_{o2} = V_1 - V_2 \quad i_{o2} = \frac{V_1 - V_2}{10}$$

$$= 16 \text{ V}$$

$$= 1,6 \text{ A}$$

Apply superposition

$$V_o = V_{o1} + V_{o2} \quad i_o = i_{o1} + i_{o2}$$

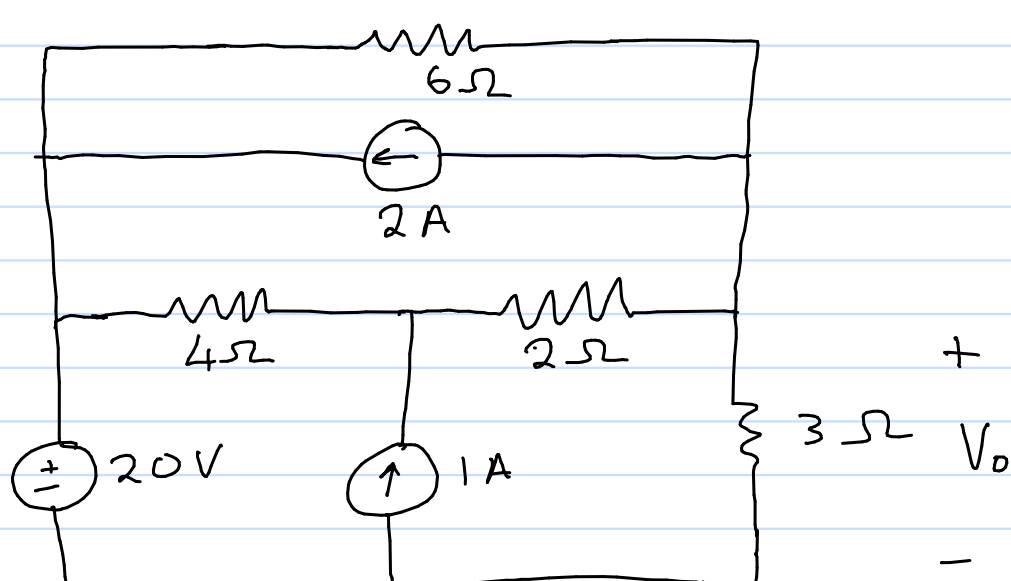
$$= 2 + 16$$

$$= 0,2 + 1,6$$

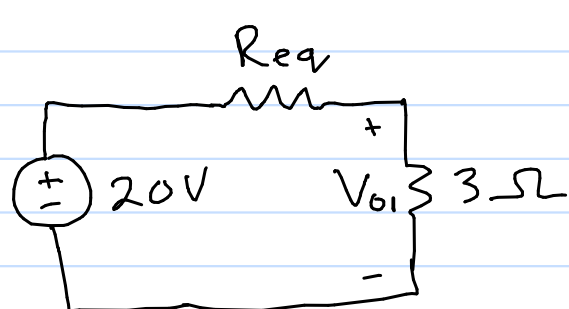
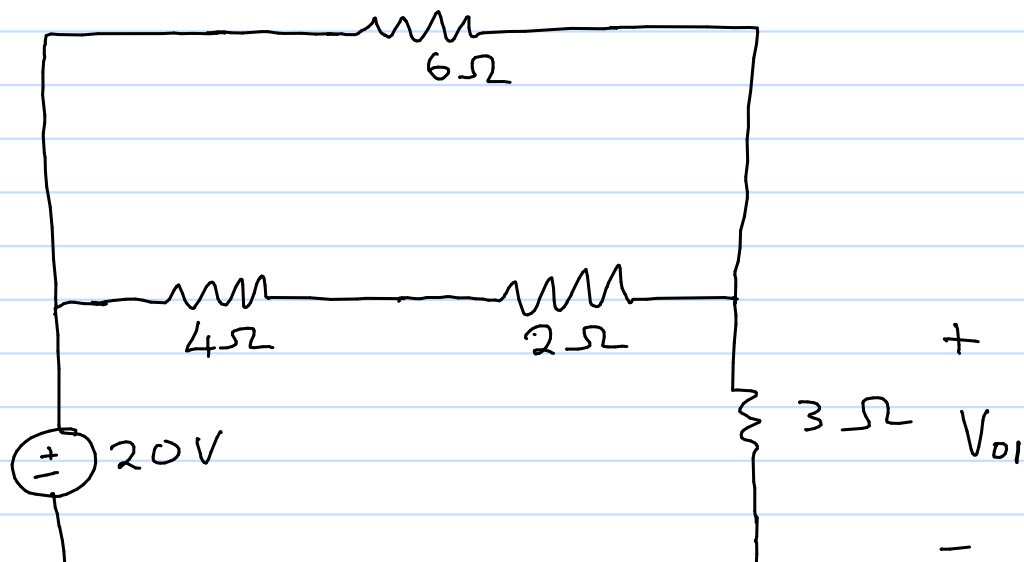
$$= 18 \text{ V}$$

$$= 1,8 \text{ A}$$

4.14 Superposition (find V_o)



Keep 20V source

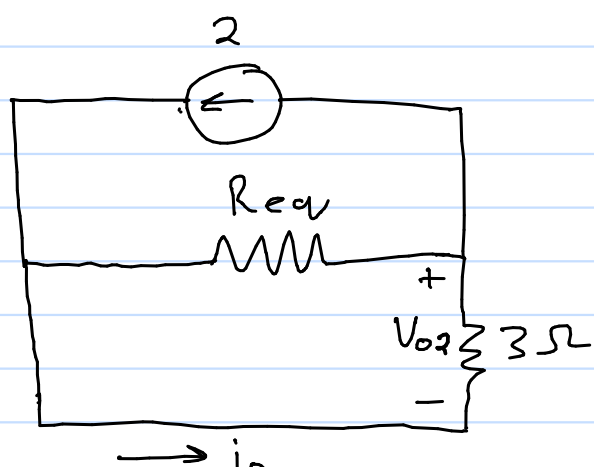
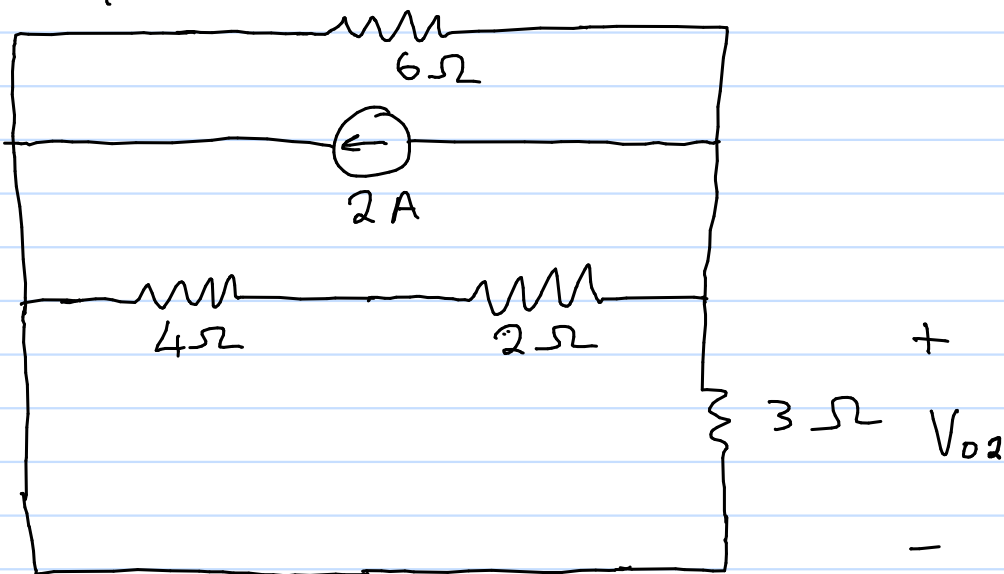


$$R_{eq} = (4+2) \parallel 6 = 3\Omega$$

Voltage division

$$V_{o1} = \frac{20 \times 3}{3+3} = 10V$$

Keep 2A source



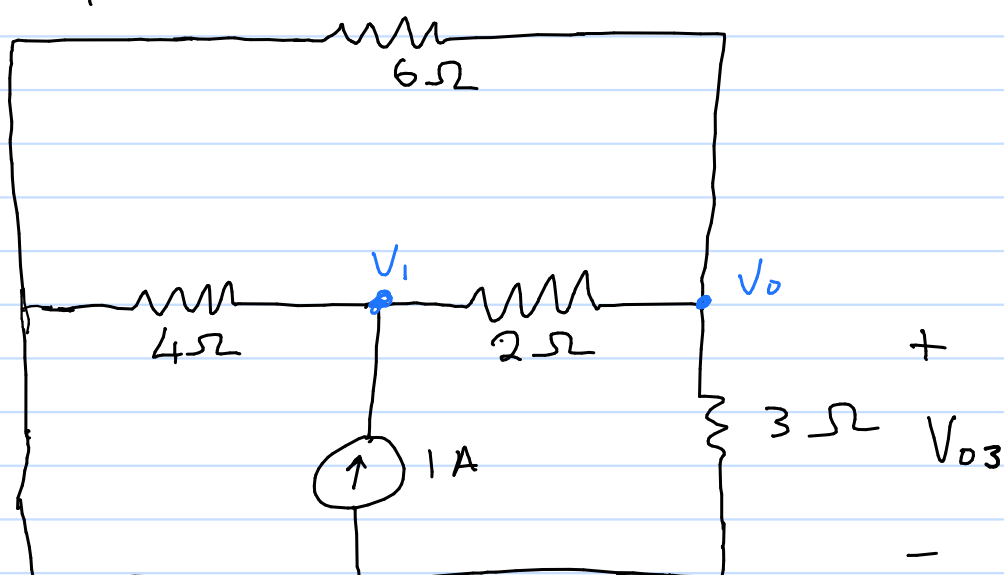
$$R_{eq} = (4+2) \parallel 6 = 3\Omega$$

Current division for i_o

$$i_o = \frac{2 \times 3}{3+3} = 1A$$

$$V_{o2} = -1 \times 3 = -3V$$

Keep 1A source



Perform nodal analysis by inspection

$$G_{00} = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} \quad G_{01} = -\frac{1}{2}$$

$$G_{10} = -\frac{1}{2} \quad G_{11} = \frac{1}{2} + \frac{1}{4}$$

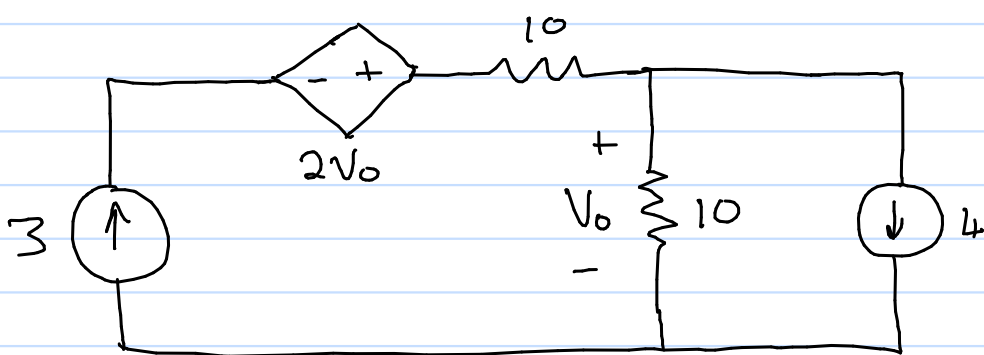
$$\begin{bmatrix} \frac{1}{1} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{4} \end{bmatrix} \begin{bmatrix} V_{o3} \\ V_1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$V_{o3} = 1, V_1 = 2$$

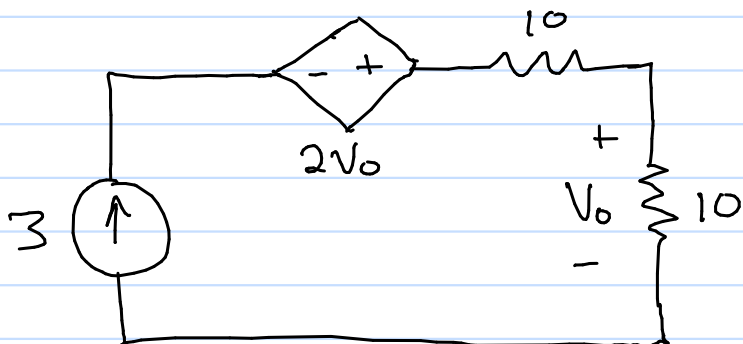
Apply superposition

$$\begin{aligned} V_o &= V_{o1} + V_{o2} + V_{o3} \\ &= 10 - 3 + 1 \\ &= 8V \end{aligned}$$

4.18 Super position (find V_o)



Keep 3A source

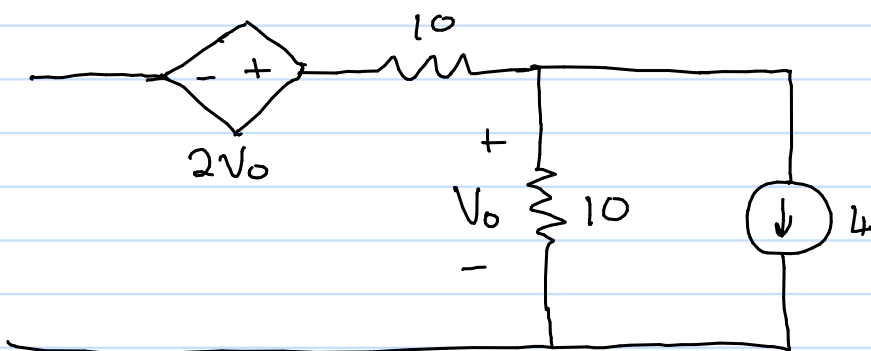


Everything is in series

Apply ohm's law

$$\begin{aligned} V_{o1} &= 3 \times 10 \\ &= 30 \text{ V} \end{aligned}$$

Keep 4A source



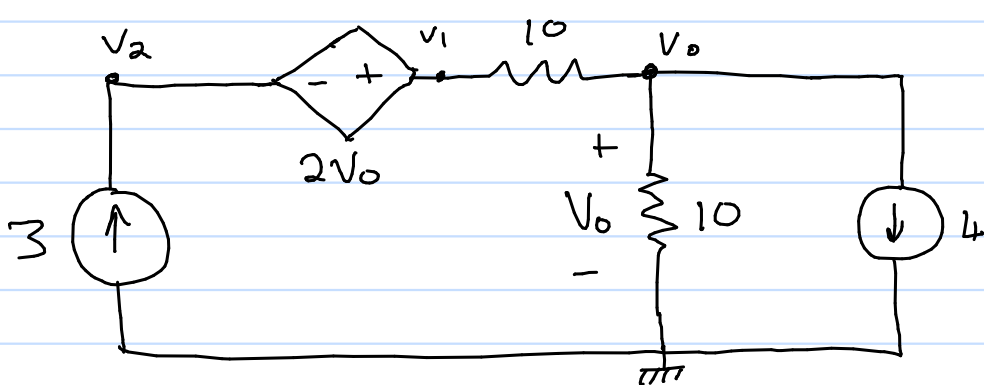
Ohm's law

$$\begin{aligned} V_{o2} &= -4 \times 10 \\ &= -40 \text{ V} \end{aligned}$$

Apply superposition

$$\begin{aligned} V_o &= V_{o1} + V_{o2} \\ &= 30 - 40 \\ &= -10 \text{ V} \end{aligned}$$

Check answer using nodal analysis



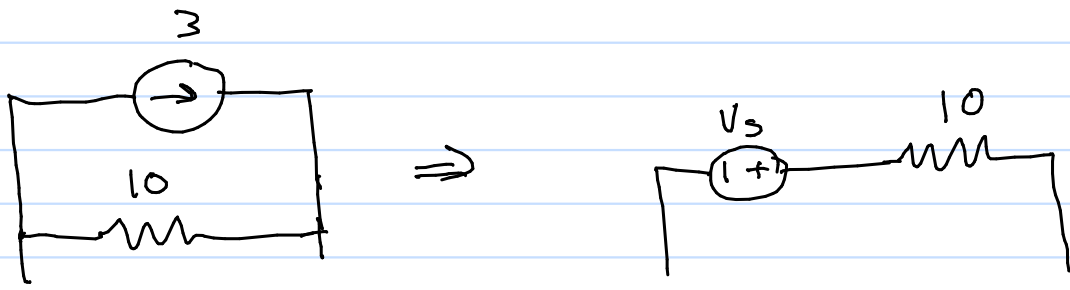
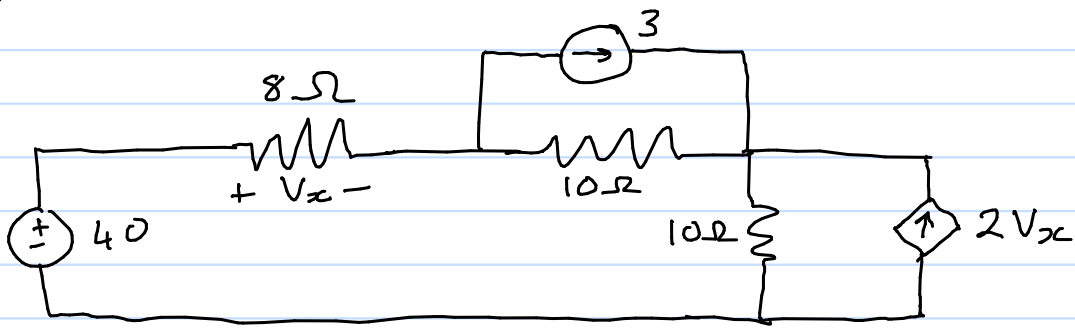
@ Node V_o

$$3 = \frac{V_o}{10} + 4$$

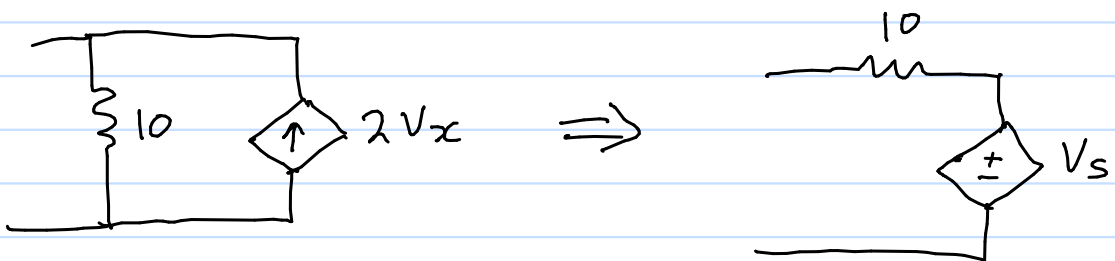
$$30 = V_o + 40$$

$$V_o = -10 \text{ V}$$

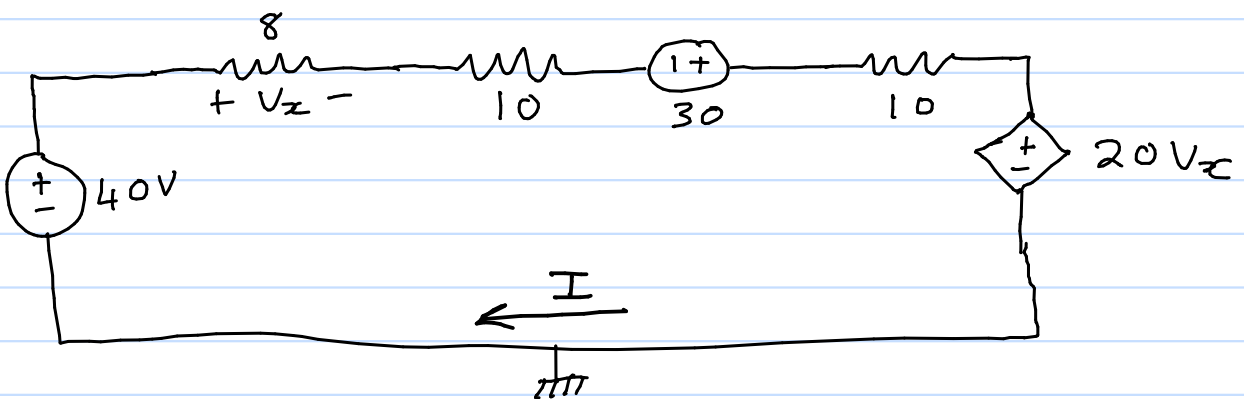
4.24 Source Transformation (find V_x)



$$\begin{aligned} V_s &= I_s R \\ &= 3 \times 10 \\ &= 30V \end{aligned}$$



$$\begin{aligned} V_s &= I_s R \\ &= 2V_x \times 10 \\ &= 20V_x \end{aligned}$$



Apply KVL

$$-40 + V_x + 10I - 30 + 10I + 20V_x = 0$$

$$21V_x + 20I = 70 \quad (1)$$

Apply Ohm's law at V_x

$$V_x = 8I$$

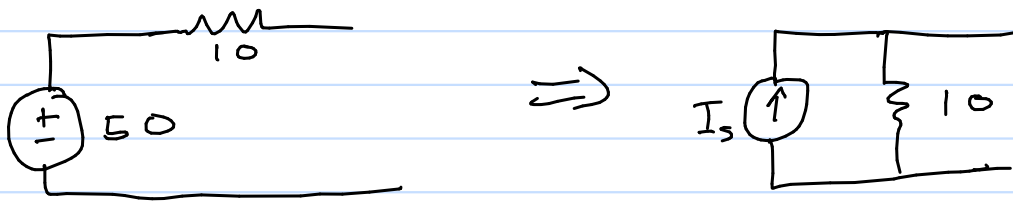
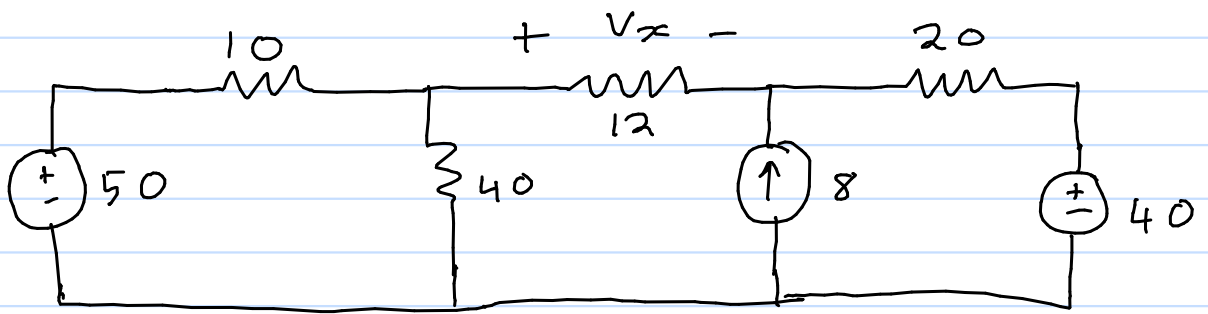
$$V_x - 8I = 0 \quad (2)$$

$$\begin{bmatrix} 21 & 20 \\ 1 & -8 \end{bmatrix} \begin{bmatrix} V_x \\ I \end{bmatrix} = \begin{bmatrix} 70 \\ 0 \end{bmatrix}$$

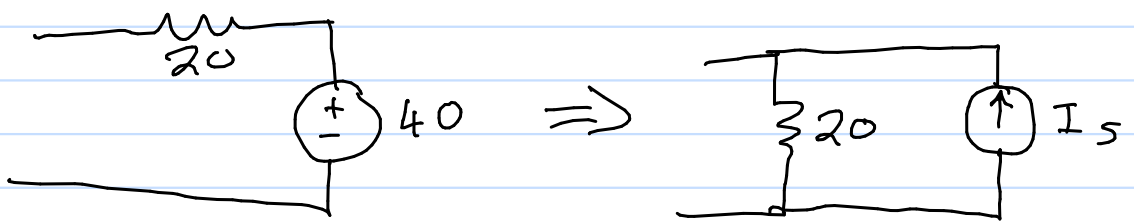
$$V_x = 2.979 \text{ V}$$

$$I = 0.372 \text{ A}$$

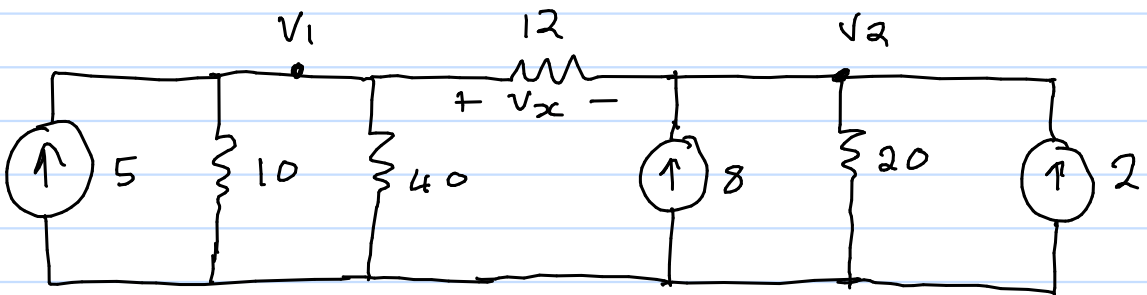
4.27 Source Transformation (find V_x)



$$\begin{aligned} I_s &= \frac{V_s}{R} \\ &= \frac{50}{10} \\ &= 5 \text{ A} \end{aligned}$$



$$\begin{aligned} I_s &= \frac{V_s}{R} \\ &= \frac{40}{20} \\ &= 2 \text{ A} \end{aligned}$$



$$V_x = V_1 - V_2$$

Nodal analysis by inspection

$$G_{11} = \frac{1}{10} + \frac{1}{40} + \frac{1}{12} \quad G_{12} = -\frac{1}{12}$$

$$G_{21} = -\frac{1}{12} \quad G_{22} = \frac{1}{12} + \frac{1}{20}$$

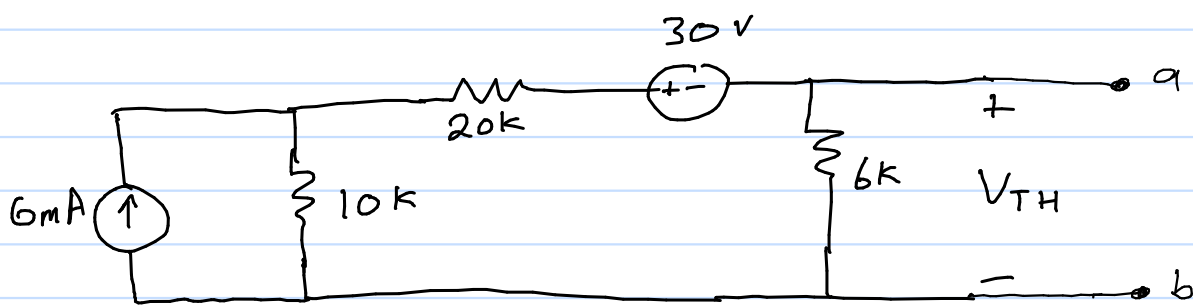
$$\begin{bmatrix} \frac{5}{24} & -\frac{1}{12} \\ -\frac{1}{12} & \frac{2}{15} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$V_1 = 72$$

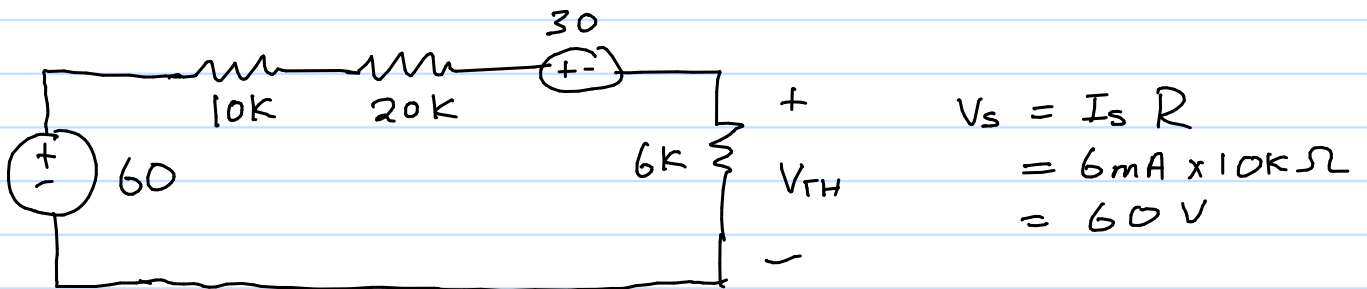
$$V_2 = 120$$

$$V_x = V_1 - V_2 = -48 \text{ V}$$

4.37 Thevenin equivalent
(across a-b)



Apply source transformation



Apply KVL

$$-60 + 30000I + 30 + V_{TH} = 0$$

$$30000I + V_{TH} = 30 \quad (1)$$

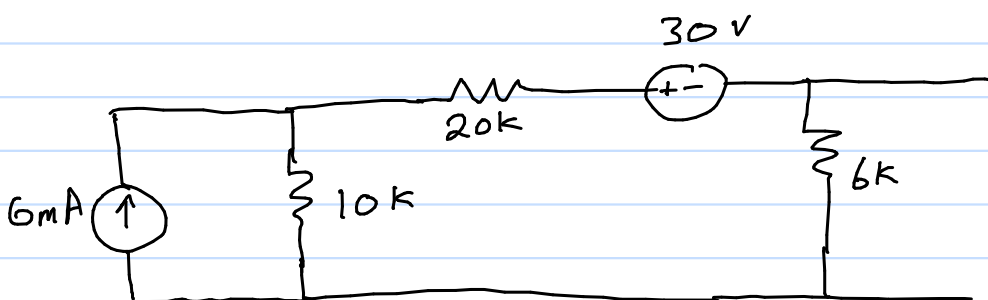
$$V_{TH} = 6000I$$

$$-6000I + V_{TH} = 0 \quad (2)$$

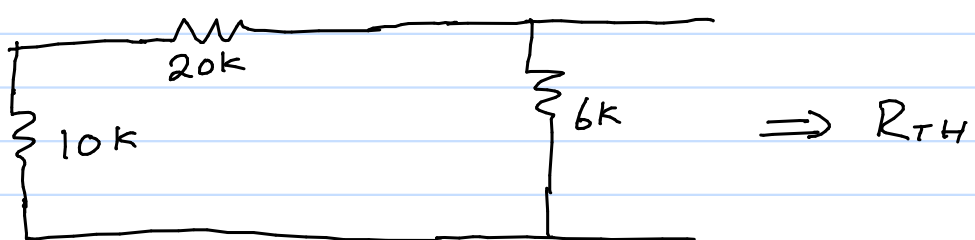
$$\begin{bmatrix} 30000 & 1 \\ -6000 & 1 \end{bmatrix} \begin{bmatrix} I \\ V_{TH} \end{bmatrix} = \begin{bmatrix} 30 \\ 0 \end{bmatrix}$$

$$I = 833.33 \mu A$$

$$V_{TH} = 5 V$$



Set all independent sources to 0

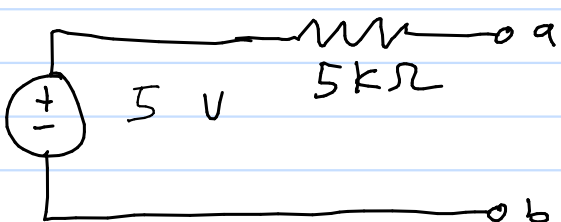


$$R_{TH} = (10k + 20k) \parallel 6k$$

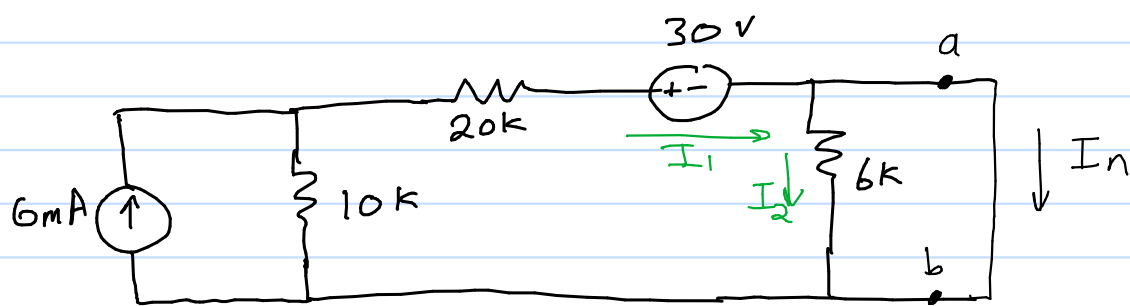
$$= \frac{6000 \times 30000}{6000 + 30000}$$

$$= 5000 \Omega$$

Equivalent circuit



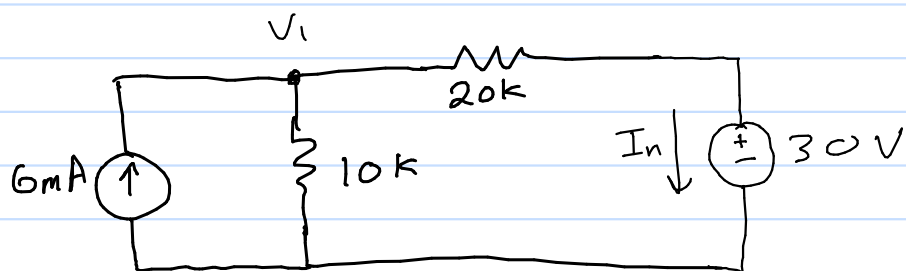
4.37 Norton Equivalent (New) (across a-b)



Through current division

$$I_2 = \frac{I_1 \times 0}{6k + 0} = 0 \quad I_n = \frac{I_1 \times 6k}{6k + 0} = I_1$$

Redraw circuit



Apply KCL at V_1

$$6 \times 10^{-3} = \frac{V_1}{10 \times 10^3} + \frac{V_1 - 30}{20 \times 10^3}$$

$$6 = \frac{V_1}{10} + \frac{V_1 - 30}{20}$$

$$120 = 2V_1 + V_1 - 30$$

$$150 = 3V_1$$

$$V_1 = 50 \text{ V}$$

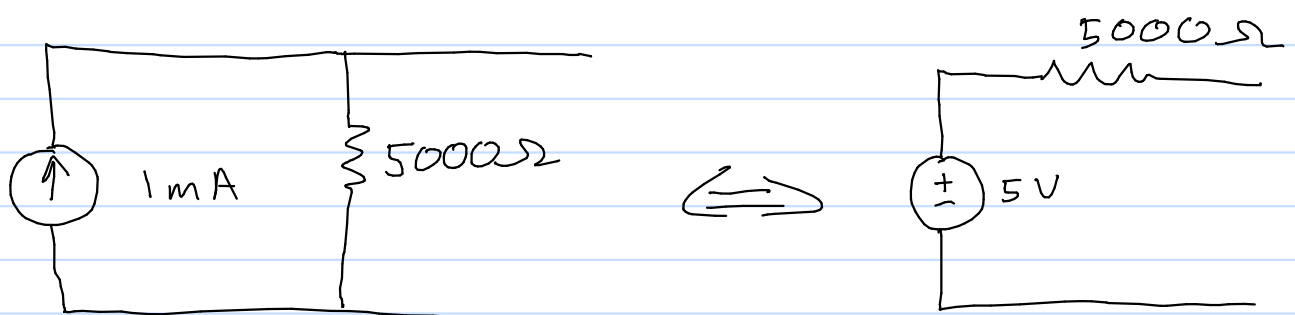
$$I_n = \frac{V_1 - 30}{20 \times 10^3}$$

$$I_n = \frac{50 - 30}{20 \times 10^3}$$

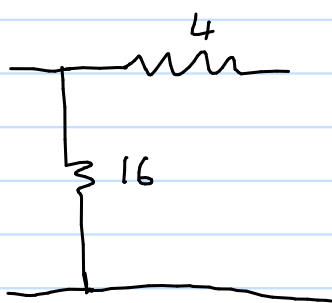
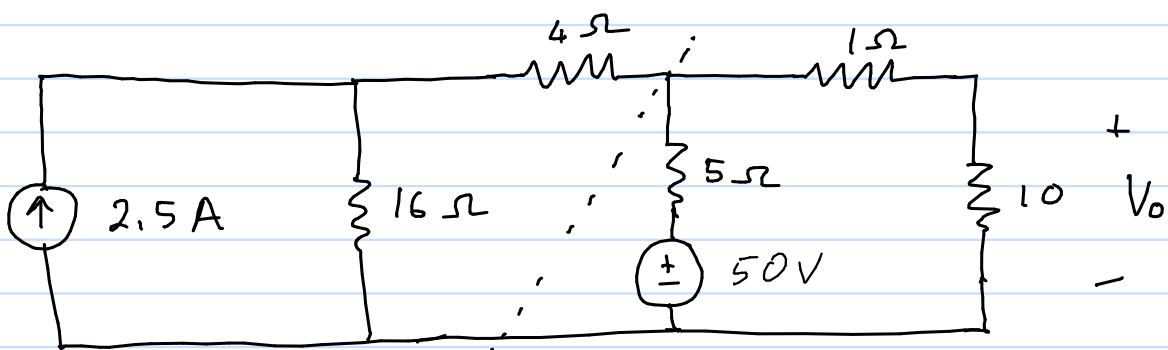
$$I_n = 1 \text{ mA}$$

$$R_N = R_{TH} = 5000 \Omega$$

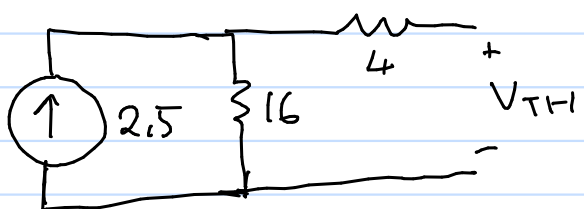
Equivalent circuit



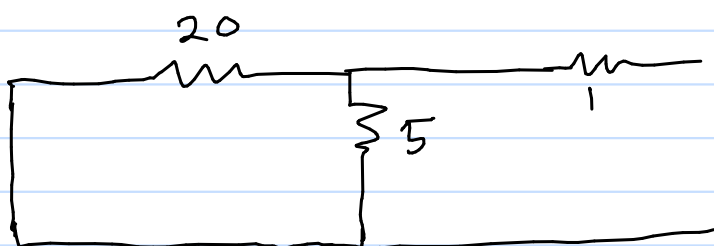
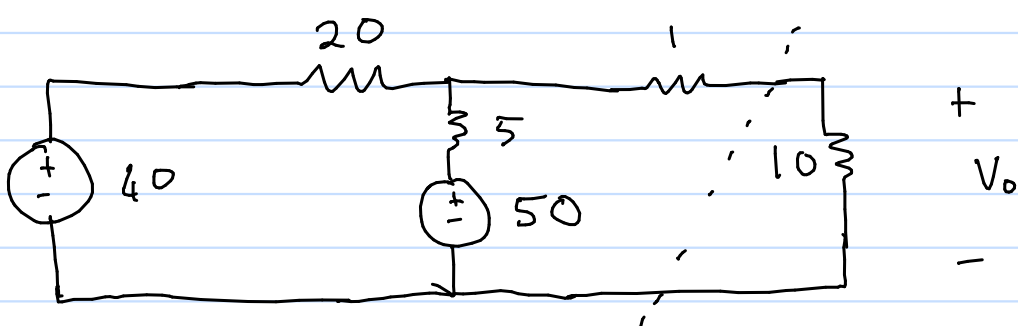
4.38 Thevenin's theorem, find V_o



$$R_{TH} = 20\Omega$$



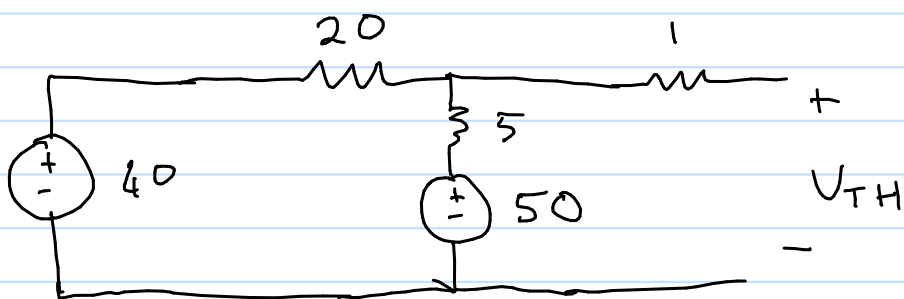
$$V_{TH} = 2.5 \times 16 = 40V$$



$$R_{TH} = 20 \parallel 5 + 1$$

$$= \frac{20 \times 5}{20 + 5} + 1$$

$$= 5\Omega$$

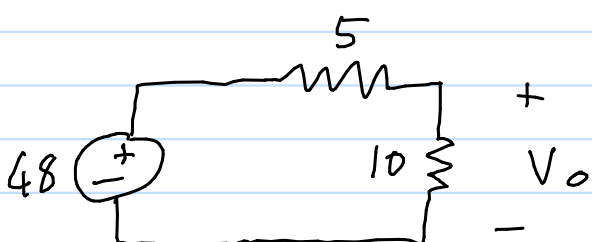


Apply KVL

$$\begin{aligned} -40 + 20I + 5I + 50 &= 0 \\ 25I &= -10 \\ I &= -0.4A \end{aligned}$$

$$\begin{aligned} -V_{TH} + 5I + 50 &= 0 \\ V_{TH} &= 50 + 5I \\ &= 50 - 5 \times 0.4 \\ &= 48V \end{aligned}$$

Find V_o

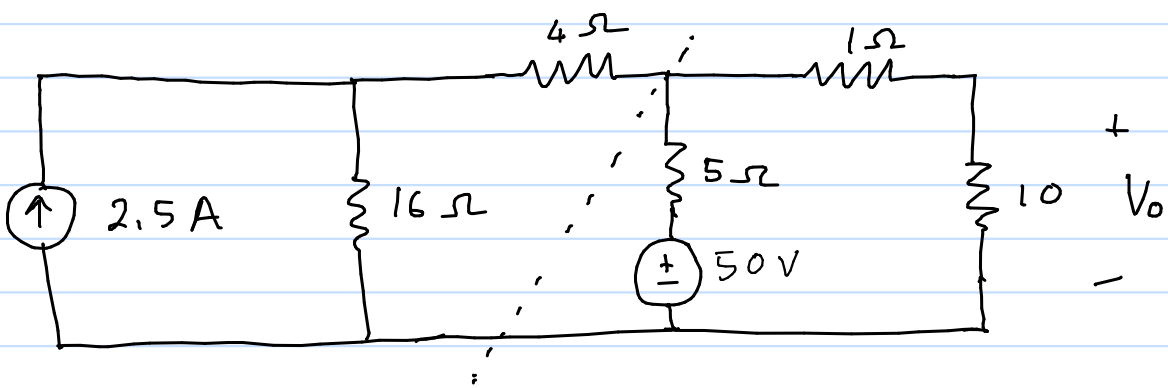


Apply voltage division

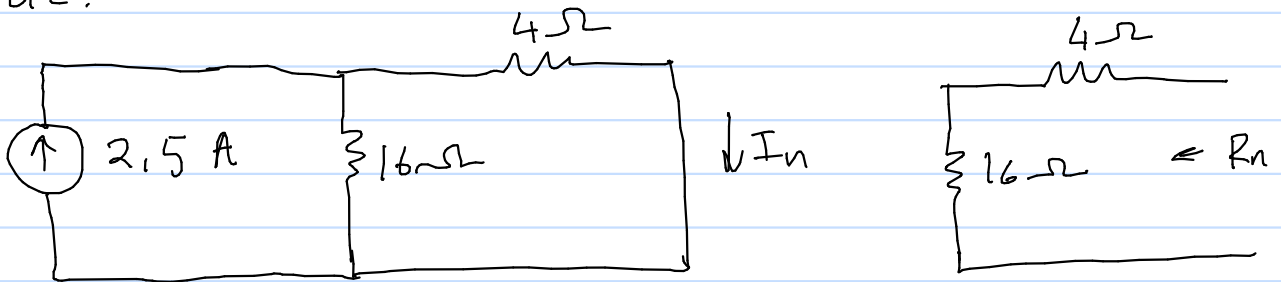
$$\begin{aligned} V_o &= \frac{48 \times 10}{5 + 10} \\ &= 32V \end{aligned}$$

Note: We have avoided nodal/mesh analysis by using Thevenin's theorem twice

4.38 Norton's theorem, find V_o (New)

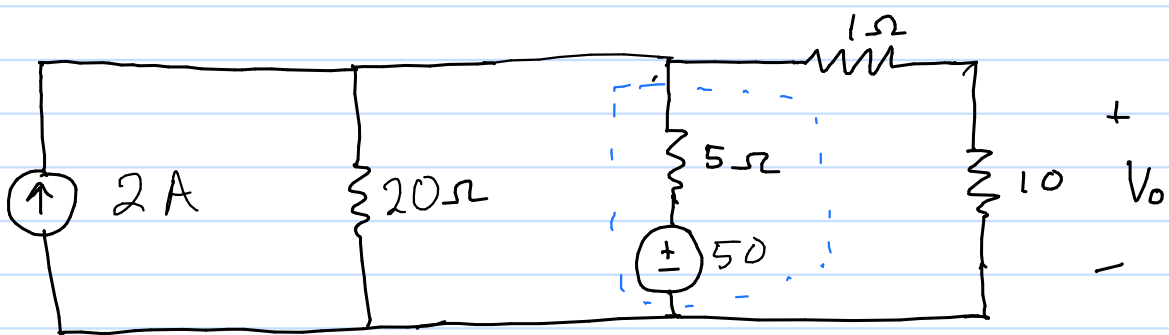


Find Norton equivalent to simplify left side.

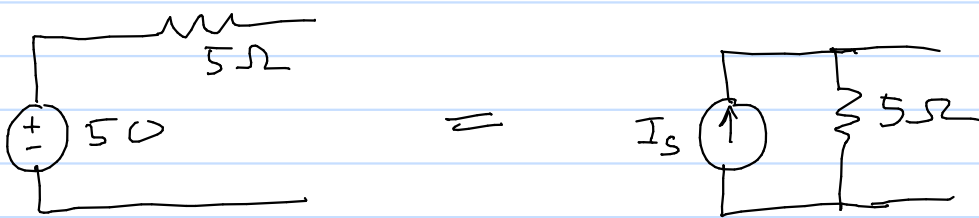


$$I_n = \frac{2.5 \times 16}{16 + 4} = 2 \text{ A} \quad (\text{Current division})$$

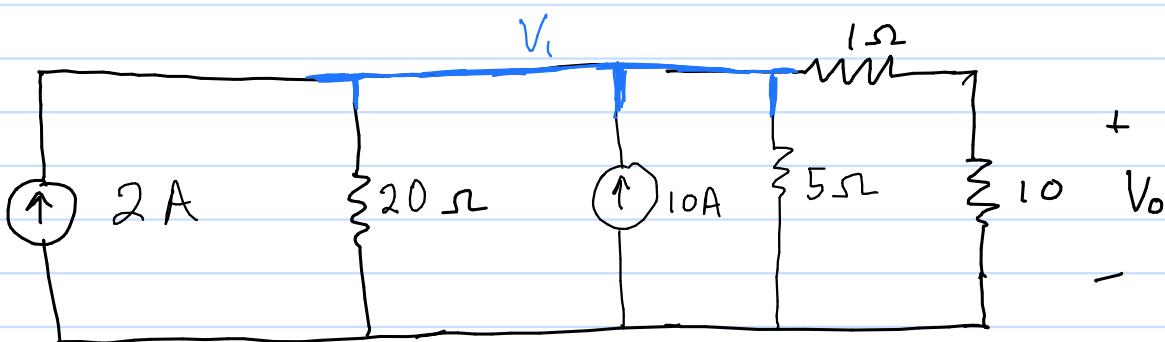
$$R_n = 16 + 4 = 20 \Omega \quad (\text{Series})$$



Source transformation



$$I_s = \frac{50}{5} = 10 \text{ A}$$



Using nodal analysis

$$\frac{V_1}{20} + \frac{V_1}{5} + \frac{V_1}{11} = 2 + 10 \quad (\times 80)$$

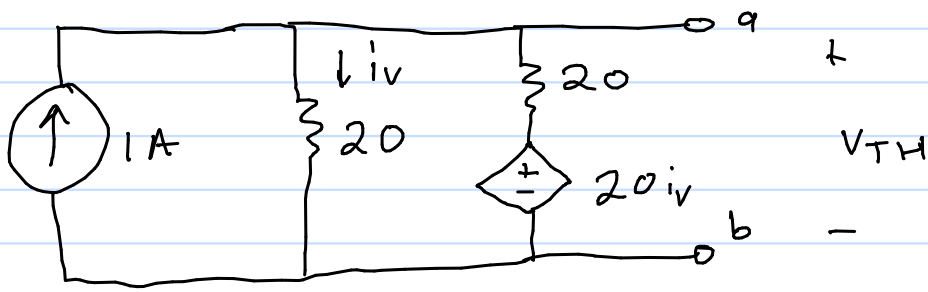
$$V_1 \left(\frac{1}{20} + \frac{1}{5} + \frac{1}{11} \right) = 12$$

$$V_1 = \left(\frac{1}{20} + \frac{1}{5} + \frac{1}{11} \right)^{-1} \times 12$$

$$V_1 = 35.2 \text{ V}$$

$$V_o = \frac{V_1 \times 10}{10 + 1} = 32 \text{ V} \quad (\text{Voltage division})$$

4.47 Thevenin equivalent (across a-b)



KCL at node V_{TH}

$$-1 + \frac{V_{TH}}{20} + \frac{V_{TH} - 20i_v}{20} = 0$$

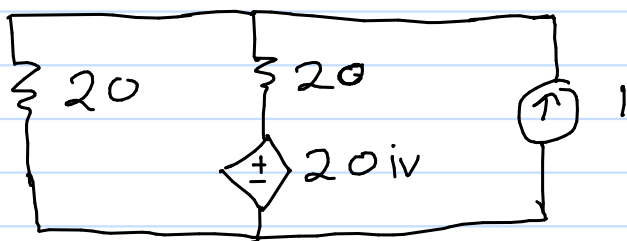
$$V_{TH} + V_{TH} - 20i_v = 20$$

$$i_v = \frac{V_{TH}}{20}$$

$$V_{TH} + V_{TH} - 20 \frac{V_{TH}}{20} = 20$$

$$V_{TH} = 20V$$

To find R_{TH} , turn off 1A source and let $i_o = 1A$



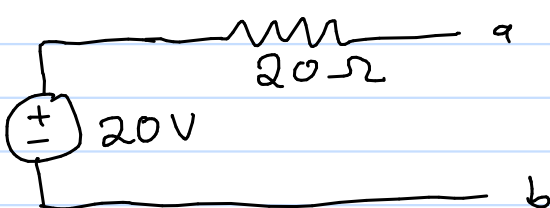
same circuit as above

$$\therefore V_o = 20V$$

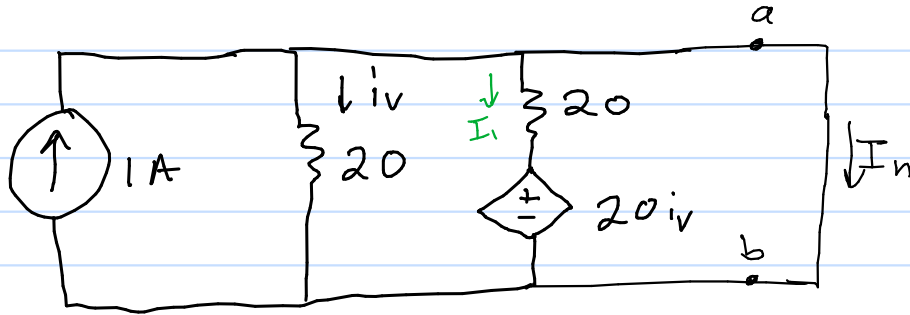
$$R_{TH} = \frac{V_o}{i_o}$$

$$= 20\Omega$$

equivalent circuit



4.47 Norton equivalent (New)
(across a-b)



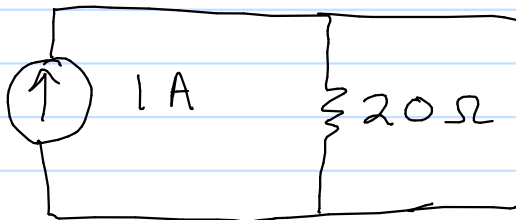
Since I_n shorts i_v and I_1

$$i_v = I_1 = 0 \text{ A}$$

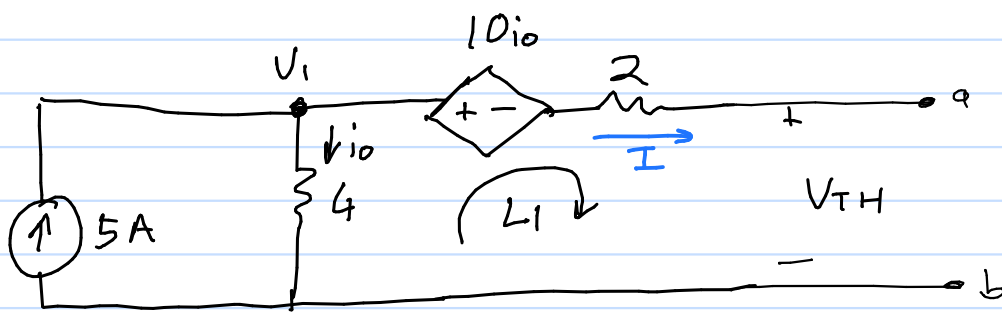
$$I_n = 1 \text{ A}$$

$$R_N = R_{TH} = 20 \Omega$$

Equivalent circuit



4.48 Thevenin equivalent (across a-b)



KVL at Loop 1

$$-V_1 + 10i_o + 2I + V_{TH} = 0 \quad (I = 0)$$

$$i_o = \frac{V_1}{4}$$

$$-V_1 + \frac{10V_1}{4} + V_{TH} = 0 \quad (\times 4)$$

$$-4V_1 + 10V_1 + 4V_{TH} = 0$$

$$6V_1 + 4V_{TH} = 0$$

KCL at node V_1

$$-5 + \frac{V_1}{4} = 0$$

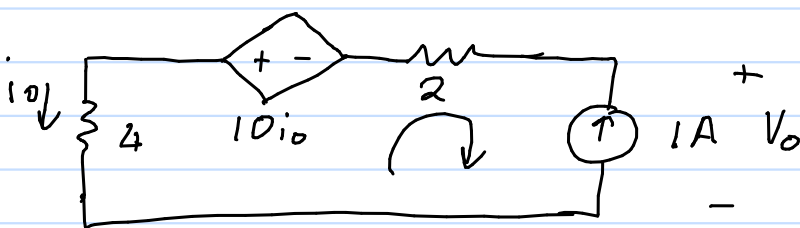
$$V_1 = 20V$$

$$6(20) + 4V_{TH} = 0$$

$$V_{TH} = \frac{-120}{4}$$

$$V_{TH} = -30V$$

Find R_{TH} , let $i_o = 1A$



$$i_o = 1A$$

Apply KVL

$$-4i_o + 10i_o - 2i_o + V_o = 0$$

$$V_o = -4i_o$$

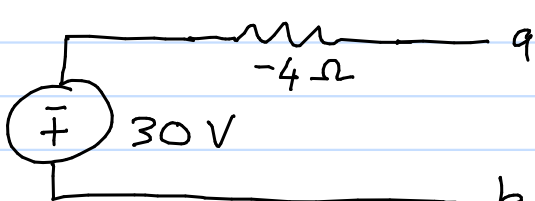
$$V_o = -4V$$

$$R_{TH} = \frac{V_o}{i_o}$$

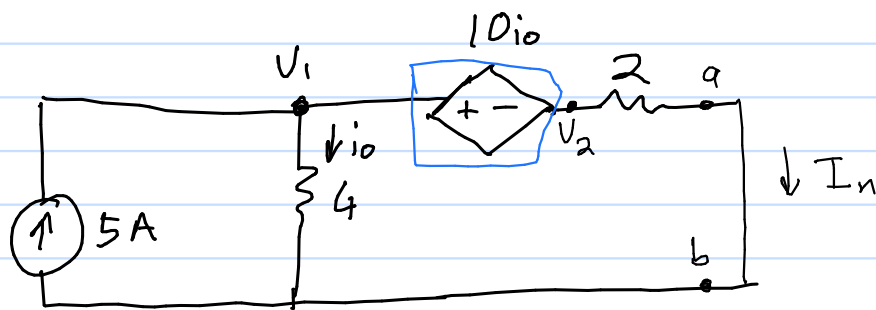
$$= \frac{-4}{1}$$

$$= -4\Omega$$

equivalent circuit



4.48 Norton equivalent (New)



KVL:

$$\begin{array}{ccc} + & + & 10i_o & - & + \\ V_1 & & V_2 & & \\ - & & - & & \end{array} \quad \begin{array}{l} -V_1 + 10i_o + V_2 = 0 \\ V_2 = V_1 - 10i_o \end{array}$$

KCL at supernode

$$i_o + I_n = 5$$

$$I_n = \frac{V_1 - 10i_o}{2}$$

$$V_1 = 4i_o$$

$$i_o + \frac{4i_o - 10i_o}{2} = 5$$

$$i_o - 3i_o = 5$$

$$i_o = \frac{5}{-2}$$

$$= -2.5 \text{ A}$$

$$I_n = \frac{4i_o - 10i_o}{2}$$

$$I_n = 7.5 \text{ A}$$

$$R_n = R_{TH} = -4 \Omega$$

Equivalent circuit

